EVIDENCE LEDGER · SIH 2026 · PS 26013

GEOVAX

Every number below is an output of one harmonisation run over real, openly licensed Indian government data. Nothing is synthetic, nothing is rounded in our favour, and the findings that make the platform look worse are in here too — because those are the ones a department would need.

Run `d41789a8-375e-4c24-9ad9-cad34853d620` **Completed** 2026-09-06 16:49 UTC

Source `out/chennai_metro/metrics.json` **Hardware** 2 vCPU · 7 GB

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01 · THE RUN

One AOI, five real layers, four minutes

Chennai — Nungambakkam · Kilpauk · Chetpet, bounded by `80.225–80.265 E`, `13.045–13.095 N`: roughly 4.3 × 5.5 km of dense central city. Deliberately modest hardware, because a platform that needs a data centre to harmonise one ward cannot be deployed to a district office.

106,077 entities harmonised	**14,614** parcels · all ULPIN-identified	**91,463** building footprints	**241.6 s** wall clock, 12 stages
20 ledger entries, chain verified	**0** synthetic features		

Where the time goes

Twelve checkpointed stages. Matching dominates, and it should — it is the only stage that reasons about 370,494 candidate pairs.

STAGE	SECONDS	SHARE	WHAT DOMINATES IT
match	141.16		two-stage feature extraction over 370,494 candidate pairs
cluster	34.93		union-find transitive closure across layers

STAGE	SECONDS	SHARE	WHAT DOMINATES IT
topology	24.72		STRtree pairwise scan and union for the gap check
resolve	11.68		Dempster–Shafer fusion over 558,294 property claims
ingest	9.87		streaming JSONL with a textual bbox pre-screen
assemble	7.61		ULPIN minting, geodesic areas, output records
reproject	4.25		one PROJ transform per geometry
structures	3.32		building-to-parcel assignment and built form
confidence	2.26		six-dimension scoring over 106,077 features
change	1.26		typed cross-source discrepancy classification
schema_map	0.55		value profiling over sampled records per layer
publish	0.00		GeoJSON, metrics, ledger, adjudication queue

02 · INGESTION

Every input is measured, not trusted

The profiler measures what arrives rather than believing the metadata that came with it. Two findings from this corpus are worth stating before anything else.

The five configured layers, as profiled in the AOI

Declared accuracy is what the publishing authority claims. It is an input to the evidence model, not a measurement — which is exactly why it is recorded.

LAYER	FEATURES IN AOI	DECLARED ACC.	COORD. DIGITS	INVALID IN SAMPLE
TNGIS cadastre · TNeGA	14,253	3.0 m	7	0

LAYER	FEATURES IN AOI	DECLARED ACC.	COORD. DIGITS	INVALID IN SAMPLE
NCSCM cadastre · MoEFCC	1,826	5.0 m	7	3
GCC buildings · municipal survey	55,810	1.0 m	7	4
Google Open Buildings v3 · extraction	70,776	1.8 m	7	0
MS Buildings TN · extraction	**0**	2.0 m	–	0

A configured layer contributed nothing, and the run says so

MS Buildings TN returned zero features in this AOI. The platform reports that as a completeness gap rather than quietly proceeding with four sources where five were configured. A pipeline that silently drops an empty layer is a pipeline whose coverage claims cannot be audited.

Precision is not accuracy, and the distinction is measured

Every vector source carries **7 decimal digits, an implied precision of 0.011 m**. A dataset can be stored to the millimetre and be three metres wrong. What this establishes is that **no source is precision-limited** — so every disagreement observed downstream is real disagreement between surveys, not rounding.

03 · ATTRIBUTE MAPPING

Four unseen schemas, crosswalked without configuration

The matcher was given four source schemas it had never seen and produced crosswalks to the canonical land-record schema with no hand-written mapping file.

SOURCE	COLUMNS	MAPPED	REPRESENTATIVE RESOLUTIONS
TNGIS cadastre	12	5	`kide` → survey_number · `lgd_village_code` → village_lgd · `sub_division` → subdivision
NCSCM cadastre	7	4	`Survey_Number` → survey_number · `Village` / `Taluk` / `District` → name fields
GCC buildings	13	6	`gcc_gis_id` → door_number · `ward_number` → ward · `road_name` → street
Google Open Buildings	4	1	`confidence` → confidence (the layer carries almost no attributes)

`kide` is the case that matters

It is the TNGIS cadastre's survey-number column. Its name conveys nothing, and no lexical matcher will ever find it. It resolves because its **values** — `437`, `438/2`, `1201/3A` — are survey numbers. Value profiling is not a refinement here; it is the only signal that works.

And the mappings it refused to apply

Candidates scoring in the middle band are surfaced for review rather than applied. This run flagged `lgd_district_code → state_lgd` at **0.544** — lexically plausible, 100% pattern match on the values, and **wrong**. A human rejects it in two seconds. An automatic mapper at a lower threshold would have corrupted the administrative hierarchy for the whole AOI.

04 · TOPOLOGY

Repairs are bounded, audited, and sometimes refused

Every geometric correction moves a boundary somebody owns, so repair is treated as a recorded act rather than a cleanup step.

LAYER	VALIDATION BEFORE REPAIR	REPAIRS	REFUSED
TNGIS cadastre	2 fatal · 150 errors · 34 warnings · planar-partition error **0.1152%** (overlap 152.2 m², gap 13,951.5 m²)	711	0
NCSCM cadastre	0 fatal · 7 errors · 13 warnings · planar-partition error **0.0497%** (gap 4,970.3 m²)	20	0

LAYER	VALIDATION BEFORE REPAIR	REPAIRS	REFUSED
GCC buildings	57 fatal · 278 errors · 817 warnings · no partition check (buildings are not a partition)	–	–
Google Open Buildings	0 fatal · 1 error · 0 warnings	–	–

Violation classes found in the TNGIS layer alone: ring orientation 14,257 · gaps 48 · slivers 59 · spikes 43 · duplicate vertices 30 · micro-areas 3 · invalid geometry 2 · multipart parcel 1. **Mean topological confidence after repair: 0.9997.**

> **The ceiling is the point**
>
> Overlaps above 5 m² and snaps beyond 0.5 m are **refused and escalated**, not absorbed. A half-metre boundary movement is a survey discrepancy, not a digitising error, and a platform that quietly closes it has just redrawn someone's parcel.

05 · REGISTRATION

Two systematic offsets nobody had declared

Before any matching, the platform estimates whether two layers are simply shifted relative to each other. Both pairs in this corpus were.

1.51 m @ 073°	3.41 m @ 309°
GCC survey ↔ Google Open Buildings from 5,226 confident pairs · residual RMS 2.51 m	TNGIS ↔ NCSCM cadastre from 469 confident pairs · residual RMS 9.05 m

Both are measured automatically, in under a second, and removed before matching. Left in, a 1.5 m offset collapses IoU for every pair simultaneously, so the matcher's most informative feature becomes noise — and downstream, every building appears to encroach slightly on its neighbour, generating thousands of false findings that each cost an officer's time to dismiss.

> **Why the median, not the mean**
>
> The sample of centroid differences is contaminated by genuine change — buildings that really did move, split or appear. The median is unmoved by up to half the sample being wrong; the mean is not. This is the difference between measuring a datum shift and averaging a datum shift together with the city's construction history.

06 · AI/ML SPATIAL MATCHING

No labelled truth set exists, so the matcher builds its own

There is no public ground-truth correspondence set for Indian cadastral layers. The matcher is therefore trained by programmatic weak supervision and then self-trained, and both stages are reported rather than folded into a single accuracy number.

GCC municipal survey ↔ Google Open Buildings extraction

The large pair: 55,810 × 70,776 possible combinations, reduced before a single feature is computed.

Possible pairs	**3.95×10^9**
Candidate pairs after blocking	350,750
Blocking reduction	**99.991%**
Weak labels generated	23,034 10,637 pos · 12,397 neg
Pairs the labelling functions abstained on	327,716
Pseudo-labels added by self-training	208,495
5-fold CV accuracy / precision / recall	0.9999 / 0.9998 / 1.0000
AUC · expected calibration error	**1.0000 · 0.0002**
Accepted matches	34,697 mean p = 0.999
Elapsed	101.1 s

Cardinality of the accepted set — what the two departments actually disagree about

RELATIONSHIP	COUNT	MEANING
1 : 1	33,723	both sources agree this is one building
1 : N	616	the extraction split what the survey holds as one structure
N : 1	408	the extraction merged what the survey holds as several
unmatched, survey only	21,415	the extraction missed these
unmatched, extraction only	35,857	not present in the municipal survey

TNGIS ↔ NCSCM cadastral compilations	
Candidate pairs	19,744
Accepted matches	1,465 · 1,338 one-to-one · 132 amalgamations
AUC · calibration error	1.0000 · 0.0002
Unmatched TNGIS parcels	12,788

Read the cross-validation figures correctly, and the low match rate as a finding

The CV numbers are measured **against the platform's own weak labels**, not against independent ground truth. They say the model reproduces the labelling functions' decision surface and generalises smoothly across it. They are **not** a claim of 100% real-world precision, and we do not present them as one.

The two cadastral compilations agree on roughly a tenth of parcels in this AOI. That is the correct answer, not a failure: one is a statewide revenue cadastre and the other a coastal-zone management compilation, produced for different purposes. **A platform reporting a high match rate here would be wrong.**

07 · CONFLICT RESOLUTION

42.7% closed automatically. The rest queued, not guessed.

558,294 property claims fused	**36,987** entities carrying a conflict	**42.7%** auto-resolved	**21,192** escalated for adjudication
839 refusals by statutory rule R-GEO-01	**2** batches the 5,000-case queue groups into		

Resolution strategies actually used: fusion on 36,148 contested properties, single-source pass-through on 522,142, unanimous agreement on 4. **Rule R-GEO-01 fired 839 times** — it refuses to fuse competing boundaries more than 10 m apart, on the grounds that a 10 m disagreement is a datum, control or identity error, and automated fusion would launder it into a confident false answer. Those cases go to a human with both boundaries and their provenance.

The escalation rate is high, and it is honest

Two thirds of entities in this AOI are known to only one source (70,486 of 106,077 single-source; corroboration rate 33.6%), and where two sources do overlap they frequently disagree beyond what their own declared accuracies justify. A platform reporting 95% auto-resolution on this corpus would be hiding disagreement, not resolving it.

The batching is what makes the remainder tractable: 5,000 queued cases group into **2 batches by cause**, so an officer settles one *kind* of disagreement at a time — roughly 125 officer-hours at 90 seconds a decision — and each decision feeds back as a training example and a Beta posterior update to that source's empirical reliability.

08 · CONFIDENCE

Nothing in central Chennai reaches grade A or B — and that is the finding

Six dimensions, scored per feature, across all 106,077

DIMENSION	MEAN
Lineage integrity	1.000
Topological	0.9997
Temporal currency	0.734
Attribute completeness	0.444
Positional	0.420
Source agreement	**0.335**
Composite	**0.587**

Grade distribution against the NAKSHA 0.5 m urban specification

GRADE	COUNT	SHARE	DISPOSITION
A — publish without review	0	0%	none reach it
B — publish, flagged	0	0%	none reach it
C — desk review	51,878	48.9%	an officer at a screen
D — field verification	54,199	51.1%	someone has to go and look

GRADE	COUNT	SHARE	DISPOSITION
E — reject	0	0%	nothing is unusable

What this actually means

The reasons are specific and measurable: the best positionally accurate source available is a **1 m municipal survey**; **66.4% of entities are known to only one source**, so their errors are undetectable by corroboration; and where two sources do exist, they disagree by more than their declared accuracies justify.

This is not a failure of the platform. It is the platform doing exactly what it was built for: quantifying, parcel by parcel and ward by ward, the gap that the NAKSHA drone survey exists to close — so survey effort can be aimed where the evidence says it is needed rather than spread uniformly. **The grade map is a plan for where to fly first.** A harmonisation platform whose first run reported that everything was fine would not be measuring anything.

09 · CHANGE AND DISCREPANCY TYPING

Cross-source mode, because these are departments and not epochs

Running contemporaneous departmental layers in temporal mode would raise a demolition notice for every building one department happens not to hold — 21,415 false mutations from one layer pair alone. Typed cross-source mode instead reports what kind of disagreement each one is.

GCC ↔ Google Open Buildings — 91,969 discrepancies, 33,530 actionable

TYPE	COUNT	READING
`SOURCE_COMMISSION`	35,857	extraction sees a building the survey does not hold
`GEOMETRIC_DISAGREEMENT`	32,512	both hold it, in different places or shapes
`SOURCE_OMISSION`	21,416	survey holds a building the extraction missed
`POSITIONAL_ONLY`	860	**suppressed** — explained by the layer-wide 3.23 m systematic component
`SUBDIVISION`	610	one structure became several
`AMALGAMATION`	408	several became one

TYPE	COUNT	READING
`NO_CHANGE`	306	agreement within tolerance

> **The 860 suppressed records are the ones that matter most**
>
> Those features moved by more than the noise threshold, but the movement is fully explained by a layer-wide systematic shift. They are typed **non-actionable** rather than raised as 860 mutations. Every one of them would otherwise have become an officer's afternoon.

10 · IDENTITY AND PROVENANCE

Identifiers that survive re-survey, and a ledger a citizen can check

PROPERTY	RESULT
ULPINs minted	**14,614**, all distinct — **2 snapped-cell collisions disambiguated** by an identity-stable nonce unique
Stability under re-survey	unchanged under shifts up to ~2 m (tested)
Checksum strength	**100% of single-character substitutions detected**, exhaustively tested over 13 positions × 31 substitutions
Ledger chain	**20 entries, chain intact** — tampering is detected at the exact entry index verified
Merkle root	`74b11afd7cc1a5f588ed6749ccf66144153e092a8ad25e94252dec8737b6cec9`
Inclusion proofs	round-trip verified for arbitrary indices; forged leaves rejected

Buildings assigned to a parcel: **56,166 (61.4%)** across 10,164 parcels. **2,961 parcels show built-up area exceeding the plot area entirely** — reported as a *data* finding before it is a compliance finding, because either the parcel boundary or the footprints are wrong, and the platform says so instead of issuing 2,961 violations.

11 · SCOPE

What is deliberately not claimed

CLAIM WE DO NOT MAKE	WHY
Real-world matcher precision	Cross-validation is against the platform's own weak labels. No independent labelled correspondence set for Indian cadastral layers exists; constructing one is the single most valuable thing a deploying department could contribute.
Positional accuracy better than the sources	The platform never invents a boundary. Its output is at best as accurate as the best observed source it selected, and it reports which one that was.
An Indian float DSM	None is openly available at the required precision. The DSM → DTM → nDSM chain is proved on a real calibrated surface model (Geobasis NRW DOM1) and labelled as such — never passed off as Indian data.
Title, ownership or encumbrance	This platform harmonises *spatial* records. A geometric confidence score must not leak into a question about who owns the land; the ULPIN genealogy exists so spatial mutation can be tracked without asserting anything about title.
An encroachment determination	Change restricted to poramboke land is reported as a *finding for verification*. Rule R-POR-01 escalates; it never determines.
A live PostGIS deployment	The full canonical schema, indexes and loaders exist. The evaluation runs file-backed on purpose, so a reviewer can reproduce it without standing up a database.

12 · REPRODUCE IT

Everything above, from a clean unpack

No figure in this ledger is transcribed by hand. Each one is read from the run record the pipeline wrote, and the ledger verifies independently of anything claimed here.

```
# the run record every number above comes from
cat out/chennai_metro/metrics.json

# verify the provenance chain and the Merkle root, independently
geovax verify --out out/chennai_metro
  -> chain intact over 20 entries
```

```
  -> 74b11afd7cc1a5f588ed6749ccf66144153e092a8ad25e94252dec8737b6cec9

# reproduce the whole run from the real corpus
make data       # fetch ~2.1 GB of real government data, clip to the AOI, checksum i
make pipeline   # the 12-stage harmonisation DAG
make test       # the test suite

# or double-click, in order
1 - Setup and Test.command  ·  4 - Run the Pipeline.command  ·  3 - Harmonisation Con
```

GEOVAX Evidence Ledger — compiled from `out/chennai_metro/metrics.json`, run d41789a8, completed 2026-09-06 16:49 UTC on 2 vCPU / 7 GB. Team Perpetually Curious · SIH 2026 · PS 26013 · Ministry of Rural Development, DoLR. Figures for the raster tier, terrain chain and null-change experiment are recorded separately in `docs/08-evaluation-results.md`.